# Classical Field Theory — Exercise Sheet 27

*Matrix compatibility, formal charges and Bäcklund kinks*

**Solutions**

The first two problems assume a smooth real KdV solution $u : I \times \mathbb{R} \to \mathbb{R}$, with $I$ an open time interval, $u_t = 6uu_x - u_{xxx}$ and spatial period $L_{\mathrm{per}} > 0$. This is a conditional hypothesis, not an existence assertion on the circle. The last problem instead uses a real field on the whole line.

**Exercise 27.1 (Converting the scalar KdV pair to matrices).** For fixed $k \in \mathbb{C} \setminus \{0\}$ consider

$$-\psi_{xx} + u\psi = k^2\psi, \qquad \psi_t = -4\psi_{xxx} + 6u\psi_x + 3u_x\psi.$$

Set $\Psi = (\psi, \psi_x)^{\mathsf{T}}$.

(a) Eliminate higher spatial derivatives to derive $\Psi_x = U\Psi$, $\Psi_t = V\Psi$. Compute every entry of $U_t - V_x + [U, V]$ for an arbitrary smooth $u$.

(b) For the assumed periodic KdV field, state exactly what Lecture 27's monodromy theorem proves for these matrices. Does it establish the Poisson commutativity or independence of the resulting traces?

**Solution.** The spectral equation gives the spatial rows immediately: $\psi_x = \Psi_2$ and $(\Psi_2)_x = (u - k^2)\Psi_1$. Its derivative is $\psi_{xxx} = u_x\psi + (u - k^2)\psi_x$. Substitution into the temporal equation gives

$$\psi_t = -u_x\psi + (2u + 4k^2)\psi_x.$$

Differentiating it in $x$ gives

$$(\psi_x)_t = [-u_{xx} + (2u + 4k^2)(u - k^2)]\psi + u_x\psi_x.$$

Thus the matrices are

$$U = \begin{pmatrix} 0 & 1 \\ u - k^2 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} -u_x & 2u + 4k^2 \\ -u_{xx} + (2u + 4k^2)(u - k^2) & u_x \end{pmatrix}.$$

For the curvature, matrix multiplication gives

$$[U, V] = \begin{pmatrix} -u_{xx} & 2u_x \\ -2(u - k^2)u_x & u_{xx} \end{pmatrix}.$$

Also

$$U_t - V_x = \begin{pmatrix} u_{xx} & -2u_x \\ u_t + u_{xxx} - (4u + 2k^2)u_x & -u_{xx} \end{pmatrix}.$$

Adding proves

$$U_t - V_x + [U, V] = \begin{pmatrix} 0 & 0 \\ u_t - 6uu_x + u_{xxx} & 0 \end{pmatrix}.$$

For the assumed KdV field it is zero. The matrices are smooth and periodic, so the period transport satisfies $T_t = [V(t, 0; k), T]$ and every $\operatorname{tr} T(t; k)^r$ is constant, for each fixed nonzero $k$ and positive integer $r$. This uses the finite-matrix transport theorem with the stated basepoint. It does not prove that traces with different labels are independent or have zero mutual field-theory Poisson brackets.

**Exercise 27.2 (Local charges from a formal Riccati series).** On a patch with $\psi \neq 0$, the scalar equation suggests $\rho = \psi_x/\psi$ and $\rho_x + \rho^2 = u - k^2$. In this problem work purely with formal Laurent series: put $\zeta = 2ik$ and seek

$$\rho = \frac{\zeta}{2} + \sum_{j\geq 1} a_j[u]\zeta^{-j}.$$

Equality means equality of each coefficient; convergence and a globally nonvanishing actual $\psi$ are not assumed.

(a) Find the recursion and the first five coefficients. Integrate them over one spatial period and identify the first three nonzero integrals.

(b) Prove the formal identity $\rho_t = \partial_x[-u_x+(2u+4k^2)\rho]$ under KdV. Conclude coefficientwise conservation. Explain which analytic conclusion about an actual spectral expansion has not been proved.

**Solution.** Put $w = \rho - \zeta/2$. Since $k^2 = -\zeta^2/4$, the Riccati equation becomes $w_x + \zeta w + w^2 = u$. Its constant coefficient gives $a_1 = u$. The coefficient of $\zeta^{-n}$ gives

$$a_{n+1} = -a_n' - \sum_{\substack{i+j=n \\ i,j\geq 1}} a_i a_j.$$

In increasing order this gives

$$\begin{aligned} &a_1 = u, \qquad a_2 = -u', \qquad a_3 = u'' - u^2, \\ &a_4 = -u''' + 4uu', \\ &a_5 = u'''' - 5(u')^2 - 6uu'' + 2u^3. \end{aligned}$$

Periodicity removes integrals of derivatives, including $a_2$ and $a_4$. It also gives $\int uu'' = -\int (u')^2$. Consequently

$$\int_0^{L_{\mathrm{per}}} a_1 = \int u, \qquad \int a_3 = -\int u^2, \qquad \int a_5 = \int [(u')^2 + 2u^3].$$

The integrals on the right are over the same period.

For the time identity put $S = -u_x + (2u+4k^2)\rho$. Differentiating the formal Riccati equation gives $(\partial_x + 2\rho)\rho_t = u_t$. A direct calculation, using $\rho_x + \rho^2 = u - k^2$, gives

$$\begin{aligned} (\partial_x + 2\rho)S_x &= -u_{xxx} + 4u_x(\rho_x + \rho^2) + (2u + 4k^2)(\rho_{xx} + 2\rho\rho_x) \\ &= -u_{xxx} + 4u_x(u - k^2) + (2u + 4k^2)u_x \\ &= 6uu_x - u_{xxx} = u_t. \end{aligned}$$

Thus $R = \rho_t - S_x$ obeys $(\partial_x + 2\rho)R = 0$. The series $R$ is bounded above in powers of $\zeta$. If its highest nonzero term were $r_m\zeta^m$, multiplication by the leading $\zeta$ in $2\rho$ would give $r_m\zeta^{m+1}$. Neither its $x$ derivative nor the negative-power coefficients can cancel that highest term. Hence no such term exists and $R = 0$ formally.

The coefficient of each power therefore gives $\partial_t a_j = \partial_x b_j$ for a periodic differential polynomial $b_j$. Integrating that finite coefficient identity proves conservation. The reasoning never interchanges an infinite series with an integral. It proves neither convergence of the Laurent series nor an asymptotic expansion of an actual monodromy eigenvalue.

**Exercise 27.3 (Bäcklund transformation and a sine-Gordon kink).** On $\mathbb{R}^{1,1}$ use $\xi = (x+t)/2$, $\eta = (x-t)/2$. For two real $C^2$ fields $\phi, \widetilde{\phi}$, put $\alpha_B = (\phi + \widetilde{\phi})/2$ and $\beta_B = (\phi - \widetilde{\phi})/2$. For a fixed $a_B > 0$, the proposed *Bäcklund system* is

$$\partial_\xi \alpha_B = a_B \sin \beta_B, \qquad \partial_\eta \beta_B = a_B^{-1} \sin \alpha_B.$$

(a) Prove that any such pair solves $\phi_{\xi\eta} = \sin\phi$ and $\widetilde{\phi}_{\xi\eta} = \sin\widetilde{\phi}$.

(b) Put $\widetilde{\phi} = 0$ and solve for the branch $0 < \phi < 2\pi$ at finite points. Convert it to physical coordinates and find its velocity.

(c) For density $L = \frac{1}{2}(\phi_t^2 - \phi_x^2) - (1 - \cos\phi)$, compute the kink charge $Q = [\phi(+\infty) - \phi(-\infty)]/(2\pi)$, energy and lower-index momentum integral $\int \phi_t \phi_x \mathrm{d}x$. This line solution is not a periodic field for the monodromy theorem.

**Solution.** Differentiation of the two first-order equations gives

$$(\alpha_B)_{\xi\eta} = \cos\beta_B \sin\alpha_B, \qquad (\beta_B)_{\xi\eta} = \cos\alpha_B \sin\beta_B.$$

Adding and subtracting proves

$$\phi_{\xi\eta} = \sin(\alpha_B + \beta_B) = \sin\phi, \qquad \widetilde{\phi}_{\xi\eta} = \sin(\alpha_B - \beta_B) = \sin\widetilde{\phi}.$$

The physical equation is $\phi_{tt} - \phi_{xx} + \sin\phi = 0$ because $\partial_\xi \partial_\eta = \partial_x^2 - \partial_t^2$.

For $\widetilde{\phi} = 0$ both half fields equal $\phi/2$. The first-order system reads $\phi_\xi = 2a_B \sin(\phi/2)$ and $\phi_\eta = 2a_B^{-1}\sin(\phi/2)$. On the specified open branch, $\mathrm{d}\log\tan(\phi/4)/\mathrm{d}\phi = [2\sin(\phi/2)]^{-1}$. Thus

$$\log\tan(\phi/4) = a_B\xi + a_B^{-1}\eta + \delta.$$

Let $\gamma = (a_B + a_B^{-1})/2$ and $v_{\mathrm{tr}} = (1 - a_B^2)/(1 + a_B^2)$. Then $\gamma > 0$, $|v_{\mathrm{tr}}| < 1$, $\gamma^2(1 - v_{\mathrm{tr}}^2) = 1$, and choosing $\delta = -\gamma x_0$ gives

$$\phi(t,x) = 4\arctan e^{\gamma(x - v_{\mathrm{tr}}t - x_0)}.$$

It remains strictly between zero and $2\pi$ at finite points and has those two limits at the spatial ends, so $Q = 1$.

Write $b_{\mathrm{SG}}(z) = 4\arctan e^z$ for its rest profile. Direct differentiation and the half-angle identities give $b'_{\mathrm{SG}} = 2\,\mathrm{sech}\, z$ and $1 - \cos b_{\mathrm{SG}} = 2\,\mathrm{sech}^2 z = (b'_{\mathrm{SG}})^2/2$. The derivative-squared integral is $\int (b'_{\mathrm{SG}})^2 \mathrm{d}z = 4\int \mathrm{sech}^2 z \mathrm{d}z = 8$. For $z = \gamma(x - v_{\mathrm{tr}}t - x_0)$, $\phi_x = \gamma b'_{\mathrm{SG}}$ and $\phi_t = -\gamma v_{\mathrm{tr}} b'_{\mathrm{SG}}$. Consequently

$$E_{\mathrm{en}} = \frac{\gamma^2(1 + v_{\mathrm{tr}}^2) + 1}{2\gamma} \int (b'_{\mathrm{SG}})^2 \mathrm{d}z = 8\gamma.$$

The equality uses $1 = \gamma^2(1 - v_{\mathrm{tr}}^2)$. The lower-index momentum integral is

$$\int \phi_t \phi_x \mathrm{d}x = -\gamma v_{\mathrm{tr}} \int (b'_{\mathrm{SG}})^2 \mathrm{d}z = -8\gamma v_{\mathrm{tr}}.$$

Its sign differs from the raised spatial momentum. The sine-Gordon potential and energy are different from the $\phi^4$ kink's potential and energy. The solution is on the line and connects distinct vacua, so the periodic monodromy conclusion has not been applied to it.